

ROAD-07-AGENT-PRODUCT-MAPPING

ROAD-07: AGENT-PRODUCT MAPPING

Carrington Storm Motors — Project AEGIS

Complete Mapping of 18 Agents of Aegis to 50+ CSM Products

Document Status: Investor-Grade Strategic Reference
Version: 1.0 — 15 July 2026
Classification: CONFIDENTIAL — Accredited Investor Eyes Only
Author: Zirconia, Director of Test & Validation — Aegis Operations Command
Based On: AGENTS_OF_AEGIS_COMPLETE.md (CSMGen), BOM-REGISTRY (CSMFAB), CSMTTest README

AGENT-PRODUCT MAPPING PHILOSOPHY

Each of the 18 Agents of Aegis is not merely a brand ambassador but the literal embodiment of a CSM product line — the engineer, designer, field operator, or researcher most intimately connected with that product’s development. The agent-product mapping serves three strategic functions: (1) it assigns clear technical ownership for each milestone in ROAD-01 and ROAD-10; (2) it provides investors with a memorable human narrative for each product’s development journey; and (3) it structures the organizational accountability that will scale from 18 agents today to 1,500+ employees at full commercial maturity. Every agent named below is the accountable party for their product’s TRL advancement, pilot program execution, and commercial launch.

AGENT-PRODUCT MAP — COMPLETE ROSTER

001 — DIRECTOR KAIROS STEELE → AEGIS-C COMPOSITE SHIELDING (DIELECTRIC CITADEL)

	Attribute	Detail
Agent		Director Kairos Steele
Codename		CITADEL
Analog Origin		Phil Coulson — founder of S.H.I.E.L.D., reborn as visionary of electromagnetic survival
Product		AEGIS-C COMPOSITE SHIELDING SYSTEM — The Dielectric Citadel
Product Description		7-layer ceramic composite panel (ZrB ₂ -SiC UHTC outer → MXene EMI shield → ZrO ₂ foam core → YInMn Blue pigment → FSS fractal etch → Ti ₃ AlC ₂ MAX phase bracket → PTFE/ZrO ₂ double gasket) achieving >148 dB SE across 1 kHz-10 GHz; the foundational shielding platform that enables all other CSM products
Current Development Phase		PROTOTYPE — 7-layer stack design validated at coupon level; full panel assembly in process
Next Milestone		M1.6: Aegis-C panel production line commission (Q2 2027); continuous LOM manufacturing
Funding Needed		\$3.4M (LOM line + FSS photolithography station)
Projected Market Entry		Q1 2028 (Aegis Home prototype); Q2 2029 (commercial panel sales)
		“No existing product combines UHTC ceramics, MXene EMI shielding, perovskite quantum dots, and fractal frequency-selective surfaces into a single 7-layer

Competitive Advantage Narrative	building panel. The Aegis-C is the Dielectric Citadel made physical — a wall that tells the Carrington Event ‘you shall not pass’ in the language of materials science.”
Key IP	B1: Dielectric Citadel 7-layer panel patent; A2: Discontinuous MXene tile patent; D6: LOM panel assembly patent
Revenue Projection (2030)	\$146M/year (Aegis-C panels as standalone product + embedded in Aegis Home)

002 — AGENT NYX CALDER → PHANTOM MK-1 AEGIS-HARDENED ROBOT

Attribute	Detail
Agent	Agent Nyx Calder
Codename	PHANTOM
Analog Origin	Melinda May — “The Cavalry,” silent warrior forged in post-Event extraction zones
Product	PHANTOM MK-1 AEGIS-HARDENED HUMANOID ROBOTIC PLATFORM
Product Description	Autonomous humanoid robot with 32 piezoelectric actuators, 7-layer Dielectric Citadel architecture, zero metallic/ferromagnetic components (380 component swaps), SiC/SiC actuator housings rated to 1,200°C, MXene full-body EMI shield at 92 dB, VO ₂ 4-mode electrochromic skin, LiFePO ₄ power system, PMMA POF data bus
Current Development Phase	PROTOTYPE — Phase 1 TEM cell EM testing in progress; chassis assembly complete; actuator calibration underway
Next Milestone	M4.1: Complete 5-phase CSMTesT protocol (Phases 2–5 by Q1 2027)
Funding Needed	\$1.8M (Phases 2–5 testing) + \$8.5M (10-unit pilot deployment) + \$12M (MK-2 development) = \$22.3M
Projected Market Entry	Q4 2028 (Phantom MK-1 pilot); Q4 2029 (Phantom MK-2 production)
Competitive Advantage Narrative	“The only humanoid robot capable of operating in a post-Carrington environment. Boston Dynamics robots would seize up in 90 seconds — their copper actuators, aluminum frames, and steel bearings are antennae for ground-induced currents. The Phantom was designed from first principles to be electromagnetically invisible. Every one of its 380 components was chosen for what it does NOT conduct.”
Key IP	B2: GIC-hardened robot architecture; B2 sub-claims: piezoelectric actuator array, SiC/SiC housings, PMMA POF data bus
Revenue Projection (2030)	\$23.8M/year (250 MK-2 units)

003 — AGENT SOLARA VANCE → PROTONIC COMMUNICATION NETWORK (H-FET)

Attribute	Detail
Agent	Agent Solara Vance
Codename	PROTON
Analog Origin	Daisy “Skye” Johnson / Quake — hacker turned hero who reinvented communication at the physical layer
Product	PROTONIC COMMUNICATION NETWORK — Grotthuss-Mediated H-FET Transceivers

Product Description	World’s first non-electromagnetic communication system using proton transport through Nafion 115 membranes and BaZrO ₃ solid-state conductors; H-FET logic gates with MXene gate electrodes (first Pd-free H-FET); 3.4 kHz voice, 100 baud sensor telemetry, 50 baud text; 100% piezo-powered; GIC-immune by fundamental physics (proton mass = 1,836 × electron mass)
Current Development Phase	CONCEPT → EARLY PROTOTYPE — Individual H-FET demonstrated at TRL 3; gate family TRL 2; integrated transceiver TRL 2
Next Milestone	M5.1: Protonic H-FET transceiver validation (Q2 2027); M5.5: 5-node field deployment (Q3 2027)
Funding Needed	\$30.55M total TRL advancement to TRL 9
Projected Market Entry	Q2 2031 (commercial protonic network deployment)
Competitive Advantage Narrative	“Every communication system on Earth relies on electromagnetism — and therefore every communication system on Earth is vulnerable to the next Carrington Event. The Protonic Network is the first communication paradigm in history that is immune to electromagnetic interference not through shielding but through fundamental physics. Protons are 1,836 times heavier than electrons. When the Sun’s magnetic field sweeps across the planet, it can push electrons around like leaves in a hurricane. Protons barely notice.”
Key IP	B4: Protonic transceiver patent; A5: BaZrO ₃ proton conductor; D5: H-FET gate fabrication; C5: Protonic mesh protocol
Revenue Projection (2031+)	\$3.6M/year (100 nodes deployed); scaling with infrastructure

004 — DR. THERON NASH → NEURAL-GRIP MRE STEERING WHEEL WRAP

Attribute	Detail
Agent	Dr. Theron Nash
Codename	NEURAL
Analog Origin	Leo Fitz — brilliant biomimetic interfaces engineer who designs for endothelial cells
Product	NEURAL-GRIP MRE STEERING WHEEL WRAP
Product Description	Magnetically responsive elastomer (LORD MRF-140CG) active vibration cancellation wrap achieving 96% attenuation at 125 Hz motor torque ripple; 12 KNbO ₃ -BaTiO ₃ piezoelectric discs delivering 7.83 Hz Schumann Resonance via capacitive coupling to driver palm; MXene discontinuous tile EMI shield at 92 dB; KLF2 transcription factor preservation for endothelial health
Current Development Phase	PROTOTYPE — Vibration cancellation validated in lab; Schumann coupling demonstrated; MRE formulation optimized
Next Milestone	M3.1: TEM cell testing (Q4 2026); AEC-Q200 automotive qualification (Q2 2028)
Funding Needed	\$1.2M (TRL advancement + certification)
Projected Market Entry	Q4 2028 (aftermarket); Q2 2029 (OEM factory option)
Competitive Advantage Narrative	“The steering wheel is the most intimate human-machine interface in the vehicle. It vibrates at the exact frequency (125 Hz) that downregulates KLF2 — the mechanosensitive transcription factor that protects your arteries from atherosclerosis. A conventional steering wheel is slowly damaging your cardiovascular

	system every time you drive. The Neural-Grip actively cancels every harmful frequency while entraining your nervous system to the Earth's own healing 7.83 Hz Schumann Resonance. It is the first automotive component designed to improve the driver's health."
Key IP	B7: Neural-Grip patent; capacitive Schumann coupling method; MRE + piezo integration architecture
Revenue Projection (2030)	~\$50M/year (part of Aegis Embark revenue stream)

005 — DR. LYRA KADE → CERVICAL-GUARD VISCOELASTIC HEADREST

Attribute	Detail
Agent	Dr. Lyra Kade
Codename	CERVICAL
Analog Origin	Jemma Simmons — biochemist and doctor who cares about every cell in the human body
Product	CERVICAL-GUARD VISCOELASTIC HEADREST
Product Description	Triple-mandate neuroprotection headrest: (1) active vibration isolation at 28 Hz achieving −31 dB cranium transmissibility, (2) 7.83 Hz Schumann Theta state bone conduction entrainment via temporal bone, (3) 100% non-metallic BOM for complete electromagnetic invisibility; polyimide-silica aerogel padding, BFRP shell, KNbO ₃ -BaTiO ₃ bone conduction transducers
Current Development Phase	PROTOTYPE — 28 Hz vibration isolation validated; bone conduction demonstrated; aerogel composite formulated
Next Milestone	M3.1: TEM cell testing (Q4 2026); human subjects trial for Theta entrainment (Q2 2027)
Funding Needed	\$1.4M (TRL advancement + human subjects trial + certification)
Projected Market Entry	Q4 2028
Competitive Advantage Narrative	"The 28 Hz cranium nodding resonance is the most damaging vibration your brain experiences every day. It causes visual vestibular mismatch — the disconnect between what your eyes see and what your inner ear feels that produces motion sickness, driver fatigue, and cumulative neurological degradation. The Cervical-Guard is the first headrest in history designed to not just prevent whiplash but to actively heal the driver's brain during transit."
Key IP	B8: Cervical-Guard patent; 28 Hz active damping + 7.83 Hz entrainment + non-metallic BOM
Revenue Projection (2030)	~\$45M/year (part of Aegis Embark revenue stream)

006 — COMMANDER ROOK DRAVEN → MAG-FLOAT DIAMAGNETIC SEAT RAIL

Attribute	Detail
Agent	Commander Rook Draven
Codename	MAG-FLOAT
Analog Origin	Grant Ward — tactical operations commander, security and mobility specialist
Product	MAG-FLOAT DIAMAGNETIC SEAT RAIL
Product Description	Vehicle seat adjustment mechanism using pyrolytic graphite ($\chi = -450 \times 10^{-6}$, strongest room-temperature diamagnetic) with permanent magnet track; completely eliminates metallic sliding contacts; provides magnetic

	levitation force; zero conductive path between seat and vehicle chassis for GIC isolation
Current Development Phase	PROTOTYPE — Levitation force demonstrated; load testing underway (200 kg target); positive mechanical lock designed
Next Milestone	M3.1: TEM cell testing (Q4 2026); FMVSS 207 crashworthiness testing (Q2 2028)
Funding Needed	\$980K (TRL advancement + FMVSS certification)
Projected Market Entry	Q2 2029
	“Every vehicle seat rail is a conductive bridge between the occupant and the vehicle chassis — a direct path for induced current. The Mag-Float replaces metal-on-metal sliding contact with magnetic levitation. The seat literally floats on a cushion of diamagnetic repulsion. It has no conductive path. It cannot carry current. During a Carrington Event, the driver in a Mag-Float equipped seat is electrically floating — isolated from the vehicle chassis by 450 parts per million of diamagnetic susceptibility.”
Competitive Advantage Narrative	
Key IP	B9: Mag-Float patent; diamagnetic levitation for vehicle seat application
Revenue Projection (2030)	~\$35M/year (part of Aegis Embark revenue stream)

007 — ENGINEER ORION CROSS → SILENCE-BLOCK LRAM DOOR INSERT SET

	Attribute	Detail
Agent		Engineer Orion Cross
Codename		SILENCE
Analog Origin		Alphonso “Mack” Mackenzie — mechanical engineer who builds things that last
Product		SILENCE-BLOCK LRAM (LOCALLY RESONANT ACOUSTIC METAMATERIAL) DOOR INSERT SET
Product Description		Automotive door cavity insert with 82 Hz-tuned acoustic metamaterial array achieving >18 dB transmission loss at panel coincidence frequency; BFRP frame with embedded resonator masses in viscoelastic matrix; also reduces 50 Hz power line hum and 100-200 Hz motor whine; secondary EMI attenuation via MXene coating on resonator surfaces
Current Development Phase		PROTOTYPE — 82 Hz tuning validated in impedance tube; door assembly testing underway
Next Milestone		M3.1: TEM cell testing + full door assembly acoustic testing (Q4 2026)
Funding Needed		\$850K (TRL advancement + door assembly validation)
Projected Market Entry		Q3 2028
		“Vehicle doors are the acoustic weak point of the cabin — large flat panels that resonate at their coincidence frequency and transmit road noise directly to the occupant. Conventional damping adds mass. Silence-Block uses locally resonant acoustic metamaterials — tiny tuned mass-spring systems embedded in the door cavity that absorb specific frequencies through destructive interference. It is lighter, more effective, and completely non-metallic.”
Competitive Advantage Narrative		
Key IP		B10: Silence-Block LRAM patent; 82 Hz metamaterial architecture
Revenue Projection (2030)		~\$40M/year (part of Aegis Embark revenue stream)

008 — OPERATIVE ZEPHYR ARDEN → THORAX-CALM STF SEATBELT COVER SET

Attribute	Detail
Agent	Operative Zephyr Arden
Codename	THORAX
Analog Origin	Bobbi Morse / Mockingbird — combat specialist focused on personal protection
Product	THORAX-CALM STF (SHEAR-THICKENING FLUID) SEATBELT COVER SET
Product Description	Seatbelt webbing cover with flexible bladder containing shear-thickening fluid (4 kPa·s at impact rates); increases energy absorption by >100% during rapid deceleration; reduces belt webbing load on occupant thorax; STF formulation: silica nanoparticles in polyethylene glycol carrier, tuned to transition at collision-relevant shear rates; integrated KNbO ₃ -BaTiO ₃ piezo strips for active tension monitoring
Current Development Phase	PROTOTYPE — STF formulation validated; energy absorption testing on sled completed; piezo integration underway
Next Milestone	M3.1: TEM cell testing (Q4 2026); FMVSS 209 seatbelt assembly testing (Q1 2028)
Funding Needed	\$720K (TRL advancement + FMVSS certification)
Projected Market Entry	Q3 2028
Competitive Advantage Narrative	“A seatbelt’s job is to stop you gently. A conventional belt does this through webbing stretch and a load limiter — both metal components. The Thorax-Calm adds a third mechanism: a shear-thickening fluid that is liquid at rest and solid during impact. At the moment of collision, the fluid transitions — absorbing energy by doing the work of shearing through a suddenly solid medium. It has no metal. It cannot carry current. It adds protection without adding electromagnetic vulnerability.”
Key IP	B11: Thorax-Calm STF seatbelt patent; STF formulation for automotive restraints
Revenue Projection (2030)	~\$30M/year (part of Aegis Embark revenue stream)

009 — AGENT CYPHER VOSS → STELLAR-TINT EV WINDOW FILM

Attribute	Detail
Agent	Agent Cypher Voss
Codename	STELLAR
Analog Origin	Lance Hunter — stealth and reconnaissance specialist
Product	STELLAR-TINT EV WINDOW FILM
Product Description	VO ₂ vanadium dioxide electrochromic window film with 4-mode active chromatic switching: Mode 1 Aegis Blue (NIR-reflective, 85% IR rejection for thermal management), Mode 2 Emergency Cobalt Flare Orange (thermal hazard alert + rescue beacon), Mode 3 Schumann Green 534 nm (ipRGC melanopsin stimulation for occupant calming), Mode 4 Stellar Camo (adaptive NIR scatter pattern for covert SAR avoidance); MXene EMI coating between film layers; YInMn Blue pigment enhancement
Current Development Phase	PROTOTYPE — VO ₂ electrochromic switching demonstrated in 4 modes; NIR rejection measured at 85%; MXene interlayer validated; outdoor durability

	testing initiated
Next Milestone	M3.1: TEM cell testing (Q4 2026); FMVSS 205 glazing compliance (Q1 2028); AEC-Q200 (Q2 2028)
Funding Needed	\$1.6M (TRL advancement + automotive certification)
Projected Market Entry	Q4 2028
Competitive Advantage Narrative	“Electric vehicle windows are thermal weak points — up to 60% of cabin heat gain comes through the glass. Conventional tinted windows are passive — one optical state forever. Stellar-Tint is actively switchable between four states: keep heat out (Mode 1), signal for help (Mode 2), calm the driver (Mode 3), or disappear from radar (Mode 4). It is a Faraday cage and a mood ring and a rescue beacon, all in a film thinner than a human hair.”
Key IP	B12: Stellar-Tint 4-mode electrochromic film patent; VO ₂ /EMI integration
Revenue Projection (2030)	~\$55M/year (part of Aegis Embark revenue stream; high unit price, high margin)

010 — PILOT EIRA SOLVEN → WAY-FINDER HAPTIC INSOLES

Attribute	Detail
Agent	Pilot Eira Solven
Codename	WAY-FINDER
Analog Origin	Yo-Yo Rodriguez — speedster, navigator, always finding the path forward
Product	WAY-FINDER HAPTIC INSOLES
Product Description	Footwear insole with embedded PVDF-TrFE piezoelectric transducers providing haptic directional navigation cues without visual or auditory dependency; piezoelectric energy harvesting (0.5–1.0 W/step); MXene EMI shielding; integration with Marmalade app for destination setting; designed for post-Event pedestrian evacuation when GPS/cellular/satellite are unavailable
Current Development Phase	PROTOTYPE — Haptic cue patterns validated in user studies; energy harvesting efficiency verified; Marmalade API designed
Next Milestone	TEM cell testing (Q4 2026); 50-user field trial (Q2 2027)
Funding Needed	\$680K (TRL advancement + field trial)
Projected Market Entry	Q1 2029
Competitive Advantage Narrative	“When the grid collapses, your phone’s GPS is gone. Street signs are unlit. Smoke obscures landmarks. The Way-Finder insole guides you home through haptic pulses in your feet — left foot vibrates, turn left; both feet pulse, you have arrived. It harvests its own power from your walking. It has no radio, no GPS, no battery that can die. It is the most fundamental navigation system possible: your own body, guided by the Earth’s magnetic field and pre-loaded map data.”
Key IP	B17: Way-Finder haptic insole patent; haptic navigation pattern language
Revenue Projection (2030)	~\$18M/year

011 — CARETAKER BRAN SOLARA → PET-SAFE INFRASOUND CARGO LINER

Attribute	Detail

Agent	Caretaker Bran Solara
Codename	PET-SAFE
Analog Origin	Deke Shaw — unexpected caretaker with a soft heart for the vulnerable
Product	PET-SAFE INFRASOUND CARGO LINER
Product Description	Vehicle cargo area liner integrating infrasound-absorbing metamaterial (targeting 5-20 Hz range that distresses animals) + MXene EMI shielding at 92 dB; cellulose nanocrystal aerogel padding (carbon-negative, GRAS food-safe); KNbO ₃ -BaTiO ₃ piezo elements for active vibration monitoring; reduces animal stress during transport measured via heart rate variability and cortisol
Current Development Phase	PROTOTYPE — Infrasound absorption validated; CNC aerogel formulated; MXene EMI verified; animal stress study protocol designed
Next Milestone	M3.1: TEM cell testing (Q4 2026); controlled animal stress study (Q2 2027)
Funding Needed	\$580K (TRL advancement + animal study)
Projected Market Entry	Q2 2029
Competitive Advantage Narrative	“Animals hear and feel frequencies we cannot perceive — the 5-20 Hz infrasound that precedes earthquakes, the 60 Hz hum of power lines. During a geomagnetic storm, the broadband electromagnetic noise and associated vibration create a sensory assault that terrifies animals. The Pet-Safe liner absorbs those frequencies passively through metamaterial resonance and actively shields the animal from EMI — a Faraday-lined sanctuary on four wheels.”
Key IP	B15: Pet-Safe liner patent; infrasound metamaterial + EMI shielding integration
Revenue Projection (2030)	~\$12M/year

012 — GUARDIAN THALIA ROOK → CLOUD-NEST AEROGEL CHILD SEAT BASE

Attribute	Detail
Agent	Guardian Thalia Rook
Codename	CLOUD-NEST
Analog Origin	Lincoln Campbell — protector of the innocent, willing to give everything
Product	CLOUD-NEST AEROGEL CHILD SEAT BASE
Product Description	Non-metallic child safety seat with polyimide-silica hybrid aerogel composite shell (λ =0.010 W/m·K, 180 kg/m ³); electrocaloric active cooling (no compressor, no refrigerants); MXene full EMI shield at 92 dB; PVDF-TrFE piezo elements for infant vital sign monitoring (heart rate, respiration via ballistocardiography); complete non-metallic BOM for zero GIC susceptibility; integrated with Marmalade app
Current Development Phase	PROTOTYPE — Aerogel shell fabricated; electrocaloric cooling demonstrated at lab scale; MXene shielding tested; vital sign monitoring accuracy >95% vs. medical-grade reference
Next Milestone	M3.1: TEM cell testing (Q4 2026); FMVSS 213 child restraint crash testing (Q2 2027)
Funding Needed	\$1.1M (TRL advancement + FMVSS 213 certification)
Projected Market Entry	Q1 2029
	“A child safety seat is a Faraday cage that doesn’t know

Competitive Advantage Narrative	it. Add MXene to the aerogel shell and you have 92 dB of EMI shielding — the infant inside is in an electromagnetic sanctuary. Add electrocaloric cooling and the seat actively regulates temperature without a compressor. Add piezo sensors and parents can monitor their child’s heartbeat through the Marmalade app. The Cloud-Nest is the safest place a child can be during a Carrington Event — floating in aerogel, cooled by electric fields, shielded by MXene.”
Key IP	B16: Cloud-Nest patent; aerogel + electrocaloric + EMI + vital sign integration
Revenue Projection (2030)	~\$25M/year

013 — COMMANDER AURIC VEYNE → AEGIS-ASCENSION DRONE BACKPACK + SERAPHIM FLEET

Attribute	Detail
Agent	Commander Auric Veyne
Codename	ASCENSION
Analog Origin	Holden Radcliffe — transhumanist visionary, always reaching higher
Product	AEGIS-ASCENSION DRONE BACKPACK + SERAPHIM RESCUE DRONE FLEET
Product Description	Fixed-wing rescue drone with Archimedes screw propulsion (all-weather, no traditional propeller); BFRP airframe; MXene EMI shielding; Toyota solid-state battery (1,200 Wh/kg, ceramic electrolyte, zero flammable liquid); 90-minute endurance, 5 kg payload, 75 km range; Seraphim swarm coordination software for multi-drone search and rescue; BFRP backpack launcher for single-operator deployment
Current Development Phase	CONCEPT → EARLY PROTOTYPE — Archimedes screw propulsion physics validated (CSMFAB0109); airframe design complete; solid-state battery integration conceptual
Next Milestone	M4.2: Prototype first flight (Q1 2027)
Funding Needed	\$1.2M (prototype + flight testing) + \$4.5M (production line Phase II) = \$5.7M
Projected Market Entry	Q2 2029 (500 units/year Phase II) “Conventional drones use propellers — delicate, weather-sensitive, and useless in rain or debris. The Ascension uses an Archimedes screw — a helical rotor enclosed in a duct that works in rain, snow, dust, and electromagnetic hell. When the grid is down and GPS is gone, the Ascension flies pre-loaded survey patterns, identifies survivors via thermal imaging, drops supply packages, and relays communications through the Protonic Network. It is not a photography drone. It is a flying ambulance.”
Competitive Advantage Narrative	
Key IP	Archimedes screw drone propulsion; BFRP airframe; swarm coordination with protonic relay
Revenue Projection (2030)	\$14M/year

014 — OPERATIVE IRIS DORNE → ULNAR-REST CLD ARMREST

Attribute	Detail
Agent	Operative Iris Dorne
Codename	ULNAR

Analog Origin	Elena / Piper — precision operator, quiet competence
Product	ULNAR-REST CLD (CONSTRAINED-LAYER DAMPING) ARMREST
Product Description	Vehicle armrest with constrained-layer damping structure achieving loss factor $\eta > 0.35$ across 20–50 Hz; BFRP substrate with viscoelastic damping layer and cellulose nanocrystal aerogel comfort padding; reduces vibration transmission to occupant forearm by $> 80\%$; reduces ulnar nerve compression fatigue; MXene discontinuous tile EMI shielding between layers
Current Development Phase	PROTOTYPE — CLD performance validated; vibration attenuation measured; aerogel padding integrated
Next Milestone	M3.1: TEM cell testing (Q4 2026); durability cycling test (100,000 armrest deployments, Q1 2027)
Funding Needed	\$680K (TRL advancement + durability testing)
Projected Market Entry	Q1 2029
Competitive Advantage Narrative	“The armrest is the forgotten interface — your forearm rests on it for hours, absorbing 20–50 Hz vibration through the ulnar nerve into your cervical spine. The Ulnar-Rest uses constrained-layer damping — a viscoelastic layer sandwiched between two stiff layers that converts vibration into microscopic heat. It is the same physics used to quiet submarine hulls, miniaturized to fit under your elbow. And it has no metal — zero conductive path to the door frame.”
Key IP	B14: Ulnar-Rest CLD patent; BFRP + aerogel + MXene integration for armrest
Revenue Projection (2030)	~\$22M/year (part of Aegis Embark revenue stream)

015 — ENGINEER SYLAS FEN → FLUID-DAMP GEAR SHIFT INTERFACE

Attribute	Detail
Agent	Engineer Sylas Fen
Codename	FLUID-DAMP
Analog Origin	Leo Fitz / Framework Fitz — engineer of fluid and adaptive systems
Product	FLUID-DAMP MRF-BASED GEAR SHIFT INTERFACE
Product Description	Vehicle shift mechanism using LORD MRF-140CG magnetorheological fluid (80 kPa yield stress at 250 kA/m) to provide variable haptic resistance; eliminates metallic detent mechanisms; programmable shift feel (sport/comfort/eco modes via MRF field control); integrated KNbO ₃ -BaTiO ₃ piezo elements for haptic feedback and energy harvesting; BFRP housing, PEEK CF40 internal components
Current Development Phase	PROTOTYPE — MRF haptic resistance validated; programmable modes demonstrated; BFRP housing fabricated
Next Milestone	M3.1: TEM cell testing (Q4 2026); 100,000-cycle durability (Q1 2027)
Funding Needed	\$950K (TRL advancement + durability + integration testing)
Projected Market Entry	Q1 2029
Competitive Advantage Narrative	“Every gear shift mechanism uses metal detents — springs and ball bearings that click into position. Those are conductive paths, electromagnetic susceptors, and wear points. The Fluid-Damp replaces every metal component with magnetorheological fluid — a smart

	material that changes from liquid to near-solid in milliseconds when a magnetic field is applied. The shift feel is programmable — sport mode is crisp and notchy, comfort mode is smooth and effortless. Zero metal. Zero wear. Zero GIC pathway.”
Key IP	B13: Fluid-Damp MRF shift patent; MRF haptic interface architecture
Revenue Projection (2030)	~\$28M/year (part of Aegis Embark revenue stream)

016 — CAPTAIN MERIDIA VAUN → CHARLEMAGNE-CLASS FLEET RETROFIT

Attribute	Detail
Agent	Captain Meridia Vaun
Codename	CHARLEMAGNE
Analog Origin	Bobbi Morse — commanding officer, strategic leader, fleet commander
Product	CHARLEMAGNE-CLASS FLEET ELECTROMAGNETIC HARDENING RETROFIT
Product Description	Complete vessel electromagnetic hardening system for 93-vessel cruise fleet: MXene hull coating (92 dB SE, 25,000 m²/vessel), BFRP structural replacement at 200+ conductive paths, PMMA POF wiring replacement (47.3 km/vessel), LiFePO₄ battery architecture (20 MWh/vessel), FOG navigation, dielectric fire suppression, Protonic emergency communications; target <0.03% hull-induced current vs. unprotected baseline
Current Development Phase	CONCEPT → DESIGN — All 9 CSMVessel research documents complete; individual component testing at TRL 3–4; vessel-level integration design in progress
Next Milestone	M3.2: Charlemagne design freeze (Q2 2027); M3.3: Pilot vessel retrofit (Q3 2027–Q2 2029)
Funding Needed	\$47M (pilot vessel retrofit) + \$3.8M (ABS/DNV certification) = \$50.8M; total fleet program \$3.2B
Projected Market Entry	Q4 2029 (pilot vessel operational); 2030+ (fleet expansion, 6 vessels/year)
Competitive Advantage Narrative	“There are 93 cruise ships in the three major fleets — and every single one has a steel hull that acts as a 1,000-foot antenna for ground-induced currents. During a Carrington Event, a cruise ship at sea becomes a floating conductor in the Earth’s collapsing magnetic field. The Charlemagne retrofit turns that 1,000-foot antenna into a 1,000-foot insulator — MXene coating blocks EMI, BFRP breaks conductive paths, PMMA optical fiber eliminates copper wiring, and LiFePO₄ batteries provide power without creating new conductive loops. When the Sun speaks, the Charlemagne Fleet will be the only vessels at sea that can still navigate, communicate, and protect their passengers.”
Key IP	B3: Charlemagne vessel hardening patent; B5: Hexapedal amphibious mechanism; vessel-scale MXene application process
Revenue Projection (2030+)	\$320M/year (at 6 vessels/year, \$53M/vessel, decreasing with learning curve)

017 — OPERATIVE CORVUS NYX → AEGIS HOME CARINGTON STORM PROTECTION

Attribute	Detail
-----------	--------

Agent	Operative Corvus Nyx
Codename	AEGIS-HOME
Analog Origin	Grant Ward / Framework — dual nature; protector of civilian infrastructure
Product	AEGIS HOME — COMPLETE RESIDENTIAL CARRINGTON STORM PROTECTION SYSTEM
Product Description	18-product integrated residential hardening system: Aegis-C Wall Panels, BFRP Rebar, Geopolymer Concrete Foundation, Inductive-Proof Wiring (PMMA POF), Conductive-Safe Pipe System (GFRP), Safe-Wave Microwave, Stellar-Chill Refrigerator, Carrington-Proof Oven, Inflatable Ceramic Water Heater, Safe Cookware, EMF Detector, Stealth-Lift Garage Door, Dielectric Door Hinges, Non-Conductive Motor, Conductive-Proof Tools, Aerogel Thermal Breaks, Conductive-Proof Spray Cans, Dielectric Rope; target 94% GIC reduction
Current Development Phase	DESIGN COMPLETE → PROTOTYPE — All 18 BOMs finalized in BOM-REGISTRY; individual product testing at TRL 3-5; full home prototype not yet constructed
Next Milestone	M2.1: Aegis Home design freeze (Sep 2026); M2.4: Prototype construction (Q1 2027); M2.5: 50-home pilot (Q1 2028)
Funding Needed	\$2.84M (TRL advancement all products) + \$380K (prototype) + \$6.2M (50-home pilot) = \$9.42M
Projected Market Entry	Q1 2029 (volume manufacturing, 1,200 homes/year Phase II)
Competitive Advantage Narrative	“A home is the largest conductive object most people will ever own — steel rebar in the foundation, copper wiring in every wall, aluminum window frames, metal plumbing. During a Carrington Event, every one of those metal components becomes a current-carrying conductor. Aegis Home eliminates every conductive path — BFRP rebar instead of steel, PMMA optical fiber instead of copper, GFRP pipes instead of metal, ceramic hinges instead of brass. It is not a shielded home — it is a home that was never a conductor in the first place. The Dielectric Citadel Protocol applied to the place where people sleep.”
Key IP	B19: Complete non-metallic residential building system patent; 18 individual product patents
Revenue Projection (2030)	\$114M/year (1,200 homes at \$95K average)

018 — SPECIALIST AURA KIND → AEGIS EMBARK GROUNDING-SOLE FOOTWEAR		
	Attribute	Detail
Agent		Specialist Aura Kind
Codename		EMBARK
Analog Origin		Enoch the Chronicom — the outsider who understands humanity better than humans do
Product		AEGIS EMBARK GROUNDING-SOLE FOOTWEAR
Product Description		Footwear with controlled-resistivity CNT-polymer sole (10 ⁶ -10 ⁸ Ω/sq) for safe body charge dissipation while maintaining GIC isolation; KNbO ₃ -BaTiO ₃ piezoelectric energy harvesting in heel; MXene EMI shielding insole; polyimide-silica aerogel thermal insulation; PMMA POF communication thread for integration with Way-Finder and Marmalade
Current Development Phase		DESIGN → EARLY PROTOTYPE — CNT-polymer resistivity window validated; piezo harvesting tested;

Next Milestone	sole durability cycling initiated
Funding Needed	Prototype completion (Q2 2027); 50-user comfort and durability trial (Q3 2027)
Projected Market Entry	\$580K (TRL advancement + user trial) Q2 2029
Competitive Advantage Narrative	“The human body is a semiconductor — you can accumulate static charge and discharge it through the first grounded object you touch. In a post-Event environment where every metal object may be carrying induced current, that discharge could be lethal. Aegis Embark footwear provides a controlled-resistivity path to ground — dissipating static charge safely (10 ⁶ to 10 ⁸ Ω) while maintaining GIC isolation (preventing power-frequency currents from entering the body). It harvests energy from walking. It communicates with your Marmalade app. It keeps you warm. It is the foundation of personal electromagnetic safety.”
Key IP	B18: Embark footwear patent; controlled-resistivity sole for GIC safety + static dissipation
Revenue Projection (2030)	~\$15M/year

AGENT-PRODUCT DEVELOPMENT STATUS SUMMARY

Agent	Product	Phase	TRL	Next Milestone	Date	Funding to TRL 9	Revenue 2030
Steele	Aegis-C Shielding	Prototype	5-6	Q2 2027		\$3.4M	\$146M
Calder	Phantom MK-1	Prototype	4	Q4 2026		\$25.9M	\$24M
Vance	Protonic Network	Concept/Early Proto	2-3	Q2 2027		\$30.55M	\$3.6M
Nash	Neural-Grip	Prototype	5	Q4 2026		\$1.2M	~\$50M*
Kade	Cervical-Guard	Prototype	5	Q4 2026		\$1.4M	~\$45M*
Draven	Mag-Float	Prototype	4	Q4 2026		\$980K	~\$35M*
Cross	Silence-Block	Prototype	5	Q4 2026		\$850K	~\$40M*
Arden	Thorax-Calm	Prototype	5	Q4 2026		\$720K	~\$30M*
Voss	Stellar-Tint	Prototype	5	Q4 2026		\$1.6M	~\$55M*
Solven	Way-Finder	Prototype	4	Q4 2026		\$680K	\$18M
Solara	Pet-Safe	Prototype	4	Q4 2026		\$580K	\$12M
Rook	Cloud-Nest	Prototype	5	Q4 2026		\$1.1M	\$25M
Veyne	Ascension Drone	Concept/Early Proto	3	Q1 2027		\$5.7M	\$14M
Dorne	Ulnar-Rest	Prototype	4	Q4 2026		\$680K	~\$22M*
Fen	Fluid-Damp	Prototype	4	Q4 2026		\$950K	~\$28M*
Vaun	Charlemagne Fleet	Concept/Design	3-4	Q2 2027		\$75.5M	\$320M
Nyx	Aegis Home	Design/Proto	3-5	Q3 2026		\$9.42M	\$114M
Kind	Aegis Embark	Design/Early Proto	3	Q2 2027		\$580K	\$15M

*Part of aggregate Aegis Embark revenue stream (~\$367M total at Phase II). Individual agent revenue estimates are proportional allocations.

AGENT ACCOUNTABILITY FRAMEWORK

Each agent is accountable for: 1. **TRL Advancement:** Driving their product through the TRL gates defined in ROAD-02, with monthly progress reports to Zirconia 2. **BOM Maintenance:** Keeping their product’s BOM current in the BOM-REGISTRY, including cost updates, supplier changes, and cross-pollination opportunities 3. **Pilot Execution:** Leading their product’s pilot program segment (where applicable), including partner liaison, data collection oversight, and success criteria verification 4. **IP Capture:** Documenting all patentable inventions from their product’s development for the IP pipeline (ROAD-04) 5. **Budget Management:** Managing their product’s allocated budget from the funding totals above, with quarterly variance reporting to Director Steele 6.

Competitive Intelligence: Monitoring competitors in their product domain and reporting quarterly to the full agent council

"The 18 Agents of Aegis are not mascots. They are the accountable technical leads for every product Carrington Storm Motors will ever build. When an investor asks 'who owns the MXene coating process?' the answer is Director Steele, codename CITADEL. 'Who owns the Phantom actuator calibration?' Agent Calder, codename PHANTOM. 'Who ensures the Charlemagne retrofit doesn't exceed its \$47M budget?' Captain Vaun, codename CHARLEMAGNE. The mythos is the org chart. The codename is the accountable party. The product is the evidence." — Director Kairos Steele, CITADEL

Document generated 2026-07-15. All agent-to-product assignments are definitive for current development phase. As CSM scales, agent roles will transition from hands-on technical leads to product line directors managing teams of 10-50 engineers. Revenue projections are $\pm 35\%$ at P50 confidence for early-stage products and $\pm 20\%$ for prototype-stage products. Agent codenames are registered trademarks of Aegis Operations Command — all 18 codenames to be trademarked by Q4 2026.